

Optimizing LR - Gradient Descent is trying to explain **how Linear Regression finds the best line**. The slide uses calculus, but you don't need the formulas initially. Let's make it intuitive.

In this we define an error (cost) function and then minimize that error to find the best values of **m (slope)** and **c (intercept)** for the line.

Imagine You Want to Predict House Prices

You have data:

House Size Price

1000	50L
1200	60L
1500	75L

You want a line:

$$\text{Price} = m \times \text{Size} + c$$

Where:

- **m** = slope
- **c** = intercept

The question is:

What should m and c be?

Trial 1

Suppose we guess:

$$\text{Price} = 10 \times \text{Size} + 0$$

Predictions are terrible.

Error is huge.

Trial 2

Suppose we try:

$\text{Price} = 45 \times \text{Size} + 5000$

Predictions are better.

Error is smaller.

Trial 3

Suppose:

$\text{Price} = 50 \times \text{Size} + 10000$

Predictions are even closer.

Error becomes even smaller.

What Gradient Descent Does

Instead of trying millions of combinations manually:

$m = 1$
 $m = 2$
 $m = 3$
...

Gradient Descent automatically finds the best values.

Think:

Bad Line
↓
Better Line
↓
Even Better Line
↓
Best Line

Mountain Example

Imagine you're standing on a mountain in thick fog.

You cannot see the bottom.

You check:

Left side?
Right side?

Which side goes down?

Move there.

Check again.

Move again.

Eventually:

You reach the valley

That's Gradient Descent.

In Linear Regression

The mountain is actually:

Error Surface

The height represents:

How wrong the model is

High point:

Large error

Low point:

Small error

Goal:

Find minimum error

What Are m and c?

Remember:

$$y = mx + c$$

m = slope

c = intercept

Gradient Descent keeps adjusting them:

$m = 1$
 $c = 0$

↓

$m = 2$
 $c = 5$

↓

$m = 4$
 $c = 10$

↓

$m = 5$
 $c = 12$

until error becomes minimum.

Why Differentiate?

The slide shows these scary-looking formulas because differentiation tells us:

"Which direction should I move m and c to reduce the error?"

The derivative acts like a compass.

It tells us:

Go Left

or

Go Right

to reduce the error.

That's all.

What Is the Learning Rate?

Suppose you're walking downhill.

Small Steps



Safe but slow.

Huge Steps



You may jump over the valley.

Miss the optimum point.

Medium Steps



Perfect.

This step size is called:

Learning Rate (α)

The Entire Process

Start with random m and c



Calculate predictions



Calculate error



Calculate slope (gradient)



Update m and c



Calculate new error

↓
Repeat
↓
Error becomes minimum
↓
Best-fit line found

Interview Answer

Gradient Descent is an optimization algorithm used to find the best values of the slope (m) and intercept (c) in Linear Regression. It works by repeatedly calculating the error, determining the direction in which the error decreases most rapidly using derivatives, and updating the parameters until the minimum error is reached.

One-Line Memory Trick

Linear Regression = Find the best line

Gradient Descent = Find the best line automatically